FACULTY OF COMPUTING

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SECJ 3553 - ARTIFICIAL INTELLIGENCE

SECTION 03

AI Proposal

THEME:

SMART CITY – ECO-ZONE AIR CONDITIONER

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Table of Contents

Problem Statement

In the UTM library, Perpustakaan Sultanah Zanariah, the air conditioner temperature is always fixed at around 19°C regardless of the occupancy levels of every floor and room. This has caused significant energy wastage and unnecessary operational costs, especially during off-peak hours such as festive holidays, off-examination seasons and semester breaks. As a university student, we have observed that the study area on each floor is still cooled even though there are less student present than usual.

Furthermore, the temperature in the library can be affected by weather conditions outside, like rain or even day or night, yet the air conditioner temperature output remains the same. This could cause sudden changes of temperature in the study area, making it harder for student to focus on their revision.

Lastly, all air conditioners are manual controlled, meaning library staff must went up and down floor by floor of each area and rooms to set the temperature. This is just inefficient and a waste of time. Not to mention the air conditioners in the library are not centralized and not easily accessible by the library staff, leading to more fatigue.

AI solutions

Eco-Zone Aircond is an AI-powered smart conditioning management system that is designed to automate and optimize the library's cooling environment. This solution can be achieved with the implementation of IoT devices, such as motion detectors, CO_2 sensors and smart thermostat. This will allow the system to detect real-time occupancy level and environment temperature, thus removing the need for manual air conditioner setting and adjustments.

The AI system can automatically adjust the air conditioner temperature output according to the input data retrieved from the IoT devices. For example, turning off the air conditioner in certain zones when no person is detected, or setting higher temperatures when the room is cooled. Besides, the system can predict usage patterns based on the data retrieved and pre-cool specific areas accordingly.

Moreover, the system provides a centralized control dashboard for the library staff to manage. Hence, they can use the system to monitor and adjust the air conditioners' temperature output without physically visiting each floor and room. This proposed solution will save time, energy, reduce electricity costs and satisfy every university student and library staff.

Process of Emphasis in Design Thinking

In this stage, we focus on truly understanding the people who are affected by the problem, understanding the people who are affected by the problem what they experience every day, what frustrates them, and what they need. It's about seeing things from their point of view instead of just guessing what might be wrong. By listening to their stories and paying attention to how they feel, we can find deeper insights that help us come up with real, practical solutions. This understanding will guide us in designing a system that keeps the library comfortable for everyone while still being energy-efficient and environmentally friendly.

Who are we empathizing with?

The intended target audience are both undergraduate and postgraduate students who access the library for studying, group discussions and examination preparations. They spend extended periods of time which 3 to 8 hours in various parts of the library and are affected by temperature regulation. Secondary users include library staff who work 8 to 9 hours per day and are responsible for dealing with user complaints about comfort. Next, the faculty members who use library facilities for teaching and library administrators who are in charge of operational costs, energy efficiency and sustainability.

What do they need to do?

Student must be able to sit and read for long periods of time without experiencing physical discomfort. They should be able to concentrate on reading, writing assignments, studying for exams and conduct research comfortably. The library staff must maintain the facility at comforts levels while managing operations, resolving user complaints and manually setting temperatures in different zones. Finally, the library management must balance user comfort with cost-effectiveness and meet the university's sustainability requirements.

What do they see?

Students witness other students wearing jackets and hoodies in is clearly sign of overcooling. The students identify unutilized study and group discussion rooms with full air conditioning. The staff members identify temperature complaints in their feedback systems and observe air conditioning machines running continuously in unused locations, especially during late morning sessions (8-9 AM) and late evening sessions (9 – 10 PM). The management is aware of high energy expenses caused mainly by the air conditioning system.

What do they say?

User Group	Common Statements
Students	• "I can't study for long in the library because it's too cold and my hands get numb." • "The library is freezing! I always have to wear a jacket even though it is hot outside." • "The AC is freezing even though hardly anyone is using the area."
Library Staff	• "We receive daily complaints about the temperature being too cold." • "It is difficult to perform manual checks and temperature adjustments in each zone."
Management	• "We need a smarter system that keeps users comfortable while saving energy."

What do they do?

Students often take breaks to warm up outside the library, bring extra layers of clothes expressly for library trips, and some even avoid using the library at specified hours due to discomfort. Their study flow is disrupted as they move between zones to reach more comfortable temperatures. Staff members at libraries physically monitor various zones to verify conditions, manually adjust temperatures in response to complaints, and devote time to answering input regarding temperature. They frequently forget to adjust settings as occupancy shifts, which results in ongoing discomfort or energy waste.

What do they hear?

Students are always talking about how freezing the library feels, it's kind of a running joke on campus at this point. Even when the place is almost empty, the air conditioners are still blasting loudly. Library staff deal with daily complaints about the temperature, while also facing pressure from the administration to cut costs and support the university's sustainability goals. It's frustrating because they keep hearing about energy-saving initiatives, yet the library's current practices don't really match those goals.

What do they think and feel?

User Group	Thoughts and Feelings
Students	• **Distracted** from their academic work due to physical discomfort (cold hands, shivering) • **Confused** about why such obvious energy waste occurs when the university promotes sustainability • **Concerned** about getting sick from prolonged exposure to very cold temperatures, especially after coming in from hot outdoor weather
Library Staff	• **Overwhelmed** by the constant stream of temperature complaints they cannot fully address • **Helpless** because manual control systems are inadequate for managing such a large facility efficiently • **Guilty** about energy waste they witness but cannot prevent due to system limitations
Management	• **Responsible** for meeting university sustainability goals and carbon footprint reduction targets • **Eager** for innovative solutions that can achieve both user satisfaction and cost savings • **Committed** to providing excellent facilities while being good environmental stewards

Process of Define in Design Thinking

What is Define phase?

"Define" means to define the problems found during the "Empathize" phase and turn them into a clear list of needs and requirements.

What are their needs?

Need 1: Create a Comfortable Study Environment
Students need a comfortable space to stay focused while studying. When the temperature is too cold, it becomes hard to concentrate. The system should automatically adjust the temperature based on the number of people in each zone or floor, instead of relying only on manual ON and OFF controls.

Need 2: Reduce Energy Costs
The library management wants a system that helps lower energy use. By detecting empty zones, the system can switch the air conditioner to eco mode to save electricity and reduce costs.

Need 3: Respond to Outdoor Weather Conditions
The AI should not only depend on indoor sensors but also be aware of outside weather. For example, when it rains and the outdoor temperature drops, the system should slightly increase the indoor temperature to keep students and staff comfortable while saving energy at the same time.

Task Distribution

Task	Description	PIC	Deadline
Describe the process of Emphases in DT	Problem and pain, user/stakeholder and the goal are clearly described. Who is my user? What matters to them?	1.NURUL ASYIKIN BINTI KHAIRUL ANUAR 2. AFIF SHAQIR IRFAN BIN ARQAM	28/10/2025
Describe the process of Define in DT	Covers user/stakeholder needs in-depth with details and examples. What are their needs?	CHAU YING JIA	28/10/2025
The goal of AI solution	AI solution shows large amount of original thought. Ideas are creative and inventive.	NEO LI XIN	28/10/2025
Name AI solution	Eco-Zone Air Conditioner	NEO LI XIN	28/10/2025